

Age-Stratified Effect of Rivaroxaban Monotherapy for Atrial Fibrillation in Stable Coronary Artery Disease

A Post Hoc Analysis of the AFIRE Randomized Clinical Trial

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 [Supplemental content](#)

IMPORTANCE Antithrombotic therapy is crucial for older patients with coronary artery disease (CAD) and atrial fibrillation (AF) who are at a high risk of bleeding and thrombotic events.

OBJECTIVE To examine the age-stratified effects of rivaroxaban monotherapy compared with those of rivaroxaban plus antiplatelet agent combination therapy.

DESIGN, SETTING, AND PARTICIPANTS This was a post hoc secondary analysis of the Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease (AFIRE) open-label randomized clinical trial. This was a multicenter study conducted in Japan from February 23, 2015, to July 31, 2018. Patients with AF and stable CAD who had undergone percutaneous coronary intervention or coronary artery bypass grafting 1 or more years earlier or who had angiographically confirmed CAD that did not require revascularization were enrolled. Participants were stratified into 4 groups by age (<70 years, 70-74 years, 75-79 years, and ≥80 years). Study data were analyzed from August 2024 to July 2025.

INTERVENTIONS Rivaroxaban monotherapy or rivaroxaban plus antiplatelet agent therapy.

MAIN OUTCOMES AND MEASURES The primary efficacy end point was a major adverse cardiovascular event, defined as a composite of stroke, systemic embolism, myocardial infarction, unstable angina requiring revascularization, or death from any cause. The primary safety end point was major bleeding.

RESULTS This study included a total of 2215 participants (mean [SD] age, 74.3 [8.2] years; 1751 male [79.1%]). The incidence of primary efficacy end points per patient-year for rivaroxaban monotherapy vs rivaroxaban plus antiplatelet agent therapy was 3.2% vs 4.3% (<70 years), 3.2% vs 2.8% (70-74 years), 3.8% vs 5.3% (75-79 years), and 6.2% vs 10.3% (≥80 years). The hazard ratios were 0.74 (95% CI, 0.40-1.37) for those younger than 70 years, 1.16 (95% CI, 0.55-2.45) for those aged 70 to 74 years, 0.72 (95% CI, 0.41-1.26) for those aged 75 to 79 years, and 0.61 (95% CI, 0.40-0.93) for those 80 years and older (*P* for interaction = .51). For the primary safety end points, the incidence was 0.5% vs 2.3% (<70 years), 2.2% vs 2.4% (70-74 years), 1.1% vs 2.1% (75-79 years), and 2.9% vs 4.3% (≥80 years). The hazard ratios were 0.23 (95% CI, 0.06-0.79) for those younger than 70 years, 0.91 (95% CI, 0.39-2.15) for those aged 70 to 74 years, 0.52 (95% CI, 0.19-1.42) for those aged 75 to 79 years, and 0.67 (95% CI, 0.35-1.27) for those 80 years and older (*P* for interaction = .33).

CONCLUSIONS AND RELEVANCE Results of this post hoc analysis of the AFIRE randomized clinical trial reveal that rivaroxaban monotherapy reduced the risk of major cardiovascular events and major bleeding across the broad range of age in patients with AF and stable CAD. Possible age-related differences in trends, with more pronounced efficacy in older patients and more pronounced safety in younger patients, should be considered as hypothesis generating and require further research.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT02642419](#)

JAMA Cardiol. doi:10.1001/jamacardio.2025.2611
Published online August 13, 2025.

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The incidence of atrial fibrillation (AF) is increasing in aging societies, and the direct oral anticoagulant (DOAC) is the first-line treatment of AF in contemporary practice.¹⁻⁴ Antithrombotic therapy is crucial in older patients with AF who have a high risk for bleeding and thrombotic events. In older adult patients (ie, ≥ 80 years) with non-valvular AF for whom standard-dose oral anticoagulants were unsuitable, the Edoxaban Low-Dose for Elder Care Atrial Fibrillation Patients (ELDERCARE-AF) trial⁵ showed that once-daily edoxaban, 15 mg, was superior to placebo for preventing stroke or systemic embolism, without a significant increase in the incidence of major bleeding. The Optimal Thromboprophylaxis in Elderly Chinese Patients With Atrial Fibrillation registry reported that older patients with AF and coronary artery disease (CAD) had a higher risk of thrombotic and bleeding events than older patients with AF, wherein age was one of the risk factors.⁶ Accordingly, there has been renewed interest in the age-stratified use of DOACs, particularly in older patients.

Up to 10% of patients who undergo percutaneous coronary intervention (PCI) have AF, and these patients tend to be older adults.⁷⁻¹⁰ Adequate antithrombotic secondary prevention while avoiding serious bleeding complications constitutes a dilemma when using antithrombotic therapy in these patient populations. Post-PCI bleeding is associated with increased morbidity and mortality.¹¹⁻¹³ The Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease (AFIRE) trial in Japan was the first trial, to our knowledge, to demonstrate that compared with combination therapy with rivaroxaban (a non-vitamin K antagonist oral anticoagulant) and an antiplatelet agent, rivaroxaban monotherapy was equally effective for preventing cardiovascular events and reducing bleeding risk in patients with AF and stable CAD.¹⁴ Especially in older patients with AF undergoing PCI, a detailed age-based analysis is crucial because of tolerability to antithrombotic therapy, including possible polypharmacy issues, drug adherence, and poor outcomes with thrombotic or bleeding complications. However, no age-specific studies have examined these specific patients to date.

Thus, this study aimed to examine the age-stratified efficacy and safety of rivaroxaban monotherapy vs those of rivaroxaban plus antiplatelet agent combination therapy. To achieve this goal, we undertook a post hoc analysis of the AFIRE trial data.

Methods

Participants and Procedures

The design, methods, and primary results of the AFIRE trial have been previously reported.^{14,15} The AFIRE trial was a multicenter, randomized, open-label, parallel-group trial conducted in Japan that included male and female individuals aged 20 years or older with a CHADS₂ (congestive heart failure, hypertension, age ≥ 75 years, diabetes, stroke [double weight]) score of 1 or greater, stable CAD (more than 1 year after revascularization or angiographically confirmed CAD that did not require revascularization), and AF. All participants provided written informed consent before enrollment and were of Japa-

Key Points

Question Does rivaroxaban monotherapy differ from rivaroxaban-antiplatelet therapy in age-related effects on major adverse cardiovascular or bleeding events in atrial fibrillation (AF) and stable coronary artery disease (CAD)?

Findings In this post hoc analysis of the Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease (AFIRE) randomized clinical trial data, rivaroxaban monotherapy showed consistent efficacy and safety across age groups; the net efficacy was more pronounced in older patients, whereas safety benefits were more evident in younger patients.

Meaning Results reveal that rivaroxaban monotherapy yielded a consistent net clinical benefit across all age groups, especially in older patients with AF and stable CAD.

nese race and ethnicity. The trial was designed and conducted by the executive steering committee in accordance with the principles of the Declaration of Helsinki, and the study was approved by the institutional review boards of the Tohoku University, Japan, and all participating institutions. An independent data and safety monitoring committee reviewed the data from the AFIRE trial. Participants were randomly assigned (1:1) to receive either rivaroxaban monotherapy (10 or 15 mg once daily for participants with creatinine clearance 15-49 mL/min or ≥ 50 mL/min, respectively) or rivaroxaban plus antiplatelet agent therapy (either aspirin or a P2Y₁₂ inhibitor, at the discretion of the treating physician). Participants were assessed at baseline, 6 months after enrollment, and at trial completion, with additional assessments conducted for routine clinical care as needed. The follow-up period was planned to be at least 24 months and up to 45 months.

The present study was a post hoc analysis of the AFIRE trial. Participants from the modified intention-to-treat population were stratified into 4 age categories: younger than 70 years, 70 to 74 years, 75 to 79 years, and 80 years or older. The intergroup differences in efficacy and safety end points were then determined. This study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Study Outcomes

The primary efficacy end point was a major adverse cardiovascular event (MACE), a composite of stroke, systemic embolism, myocardial infarction, unstable angina requiring revascularization, or death from any cause. The primary safety end point was major bleeding, which was defined according to the International Society on Thrombosis and Hemostasis criteria.¹⁶ The secondary end point was net MACE plus major bleeding.

Statistical Analysis

Continuous variables were expressed as mean (SD) or median and IQR. Categorical variables were presented as frequency and proportion and were compared using the Wilcoxon rank sum, χ^2 , or Fisher exact test. Cumulative event rates

Table. Age Group–Stratified Baseline Characteristics

Characteristics	No. (%)				P value
	Age <70 y (n = 585)	Age 70–74 y (n = 467)	Age 75–79 y (n = 551)	Age ≥80 y (n = 612)	
Age, mean (SD), y	63.8 (5.6)	72.1 (1.4)	77.0 (1.4)	83.6 (3.0)	NA
Sex					
Female	40 (6.8)	77 (16.5)	132 (24.0)	215 (35.1)	<.001
Male	545 (93.2)	390 (83.5)	419 (76.0)	397 (64.9)	
BMI, mean (SD) ^a	25.8 (4.0)	24.5 (3.6)	24.2 (3.4)	23.5 (3.4)	<.001
Type of atrial fibrillation					
Paroxysmal	335 (57.3)	256 (54.8)	274 (49.7)	311 (50.8)	.04
Persistent	93 (15.9)	72 (15.4)	80 (14.5)	94 (15.4)	
Permanent	157 (26.8)	139 (29.8)	197 (35.8)	207 (33.3)	
Hypertension	515 (88.0)	407 (87.2)	458 (83.1)	511 (83.5)	.04
Diabetes	285 (48.7)	215 (46.0)	222 (40.3)	205 (33.5)	<.001
Dyslipidemia	431 (73.7)	325 (69.6)	386 (70.1)	396 (64.7)	.009
History of heart failure	190 (32.5)	141 (30.2)	190 (34.5)	267 (43.7)	<.001
History of stroke	74 (12.6)	70 (15.0)	81 (14.7)	98 (16.0)	.42
History of myocardial infarction	225 (38.5)	162 (34.7)	186 (33.8)	204 (33.3)	.24
History of peripheral artery disease	21 (3.6)	21 (4.5)	45 (8.2)	52 (8.5)	<.001
Current smoking	122 (20.9)	62 (13.3)	67 (12.2)	41 (6.7)	<.001
History of PCI	400 (68.4)	339 (72.65)	398 (72.2)	427 (69.8)	.36
History of CABG	58 (9.9)	49 (10.5)	66 (12.0)	79 (12.9)	.36
CHADS2 score, mean (SD)	2.0 (1.0)	1.9 (1.0)	2.9 (1.1)	3.0 (1.1)	<.001
CHA2DS2-VASc score, mean (SD)	3.1 (1.3)	3.5 (1.2)	4.6 (1.3)	4.8 (1.3)	<.001
HAS-BLED score, mean (SD)	1.8 (0.9)	2.3 (0.7)	2.2 (0.7)	2.2 (0.7)	<.001
Creatinine clearance, mean (SD), mL/min/1.73 m ²	83.8 (27.0)	66.0 (19.9)	57.1 (16.2)	43.8 (14.2)	NA
Creatinine clearance					
<30	3 (0.5)	5 (1.1)	10 (1.8)	96 (15.7)	<.001
>30 to ≤50	34 (5.8)	73 (15.6)	172 (31.2)	314 (51.3)	
>50	509 (87.0)	362 (77.5)	345 (62.6)	169 (27.6)	

Abbreviations: BMI, body mass index; CABG, coronary artery bypass grafting; CHADS2, congestive heart failure, hypertension, age ≥75 years, diabetes, stroke (double weight); CHA2DS2-VASc, congestive heart failure, hypertension, age ≥75 years (doubled), diabetes, stroke (doubled), vascular disease, age 65–74 years, sex category (female); HAS-BLED, hypertension, abnormal kidney or liver function, stroke, bleeding history or predisposition, labile international normalized ratio (INR), elderly (age >65 years), drugs or alcohol use concomitantly; NA, not applicable; PCI, percutaneous coronary intervention.

SI conversion factor: To convert creatinine clearance to milliliters per second per meter squared, multiply by 0.0167.

^a Calculated as weight in kilograms divided by height in meters squared.

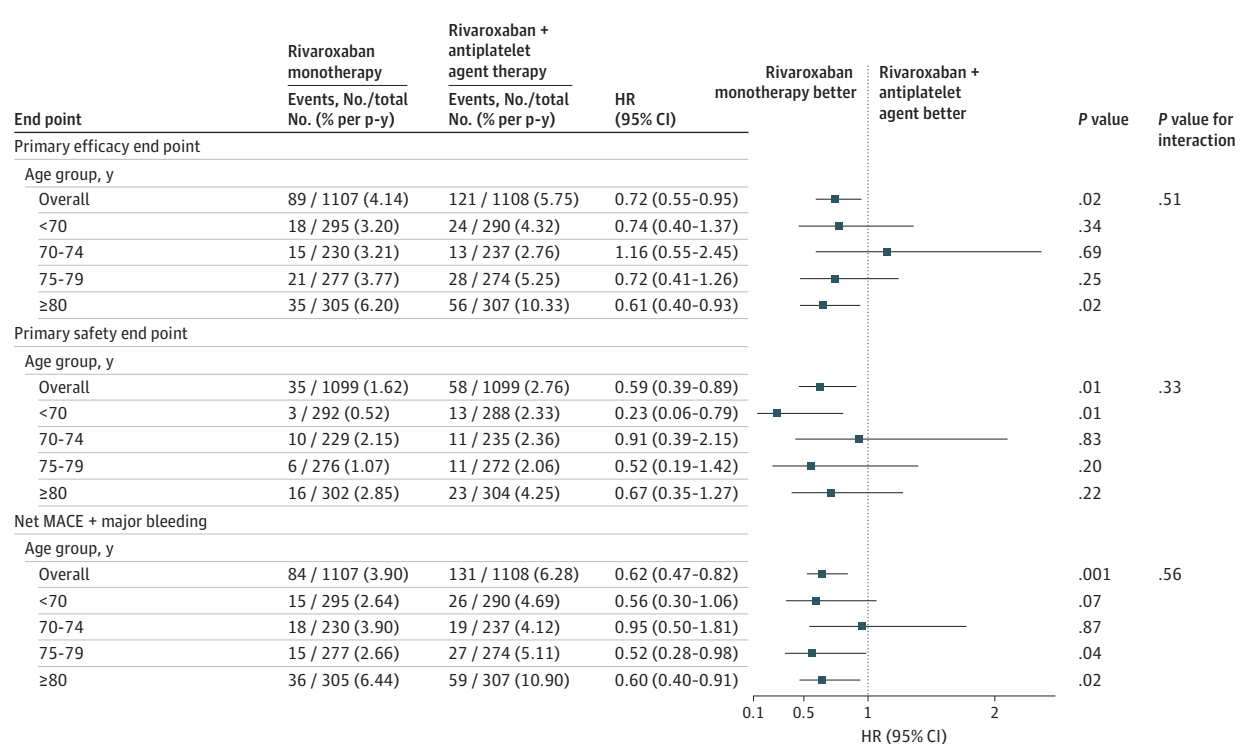
were estimated using the Kaplan-Meier method, with group-wise incidence rates expressed as percentages per patient-year. A Cox proportional hazards model was used for inter-group comparison of outcomes, and the results were expressed as hazard ratios (HRs) with 95% CIs. Using a minimization method, randomization in the AFIRE trial was adjusted for age, sex, and history of PCI, history of heart failure, hypertension, diabetes, or stroke¹⁵; therefore, the Cox hazard model was not adjusted in this analysis. The effect of rivaroxaban monotherapy on each outcome was compared with the effect of rivaroxaban plus antiplatelet agent therapy. Furthermore, the effects were examined across the age spectrum in a Cox regression model wherein age was modeled as a continuous variable. The impact of the interactions between age and treatment on efficacy and safety outcomes was tested using a logistic regression model with an interaction term between age and treatment. All statistical analyses were performed using SAS, version 9.4 (SAS Institute). Statistical significance was set at a 2-sided *P* value < .05. Study data were analyzed from August 2024 to July 2025.

Results

Participant Characteristics

In total, 2215 participants (mean [SD] age, 74.3 [8.2] years; range, 35–95 years; 464 female [20.9%]; 1751 male [79.1%]). In total, 585 participants (26.4%) were younger than 70 years (mean [SD] age, 63.8 [5.6] years), 467 (21.1%) were aged 70 to 74 years (mean [SD] age, 72.1 [1.4] years), 551 (24.9%) were aged 75 to 79 years (mean [SD] age, 77.0 [1.4] years), and 612 (27.5%) were 80 years or older (mean [SD] age, 83.6 [3.0] years). The participant characteristics are shown in the **Table**. Compared with the younger group, the older group involved more female participants; was more likely to have heart failure, peripheral artery disease, or history of coronary artery bypass grafting; had a lower body mass and lower rates of diabetes and dyslipidemia; and tended to have lower estimated glomerular filtration rates. The details of each age-stratified treatment allocation are shown in eTable 1 in **Supplement 1**.

Figure 1. Age Group–Stratified Clinical Outcomes



Primary Efficacy End Point

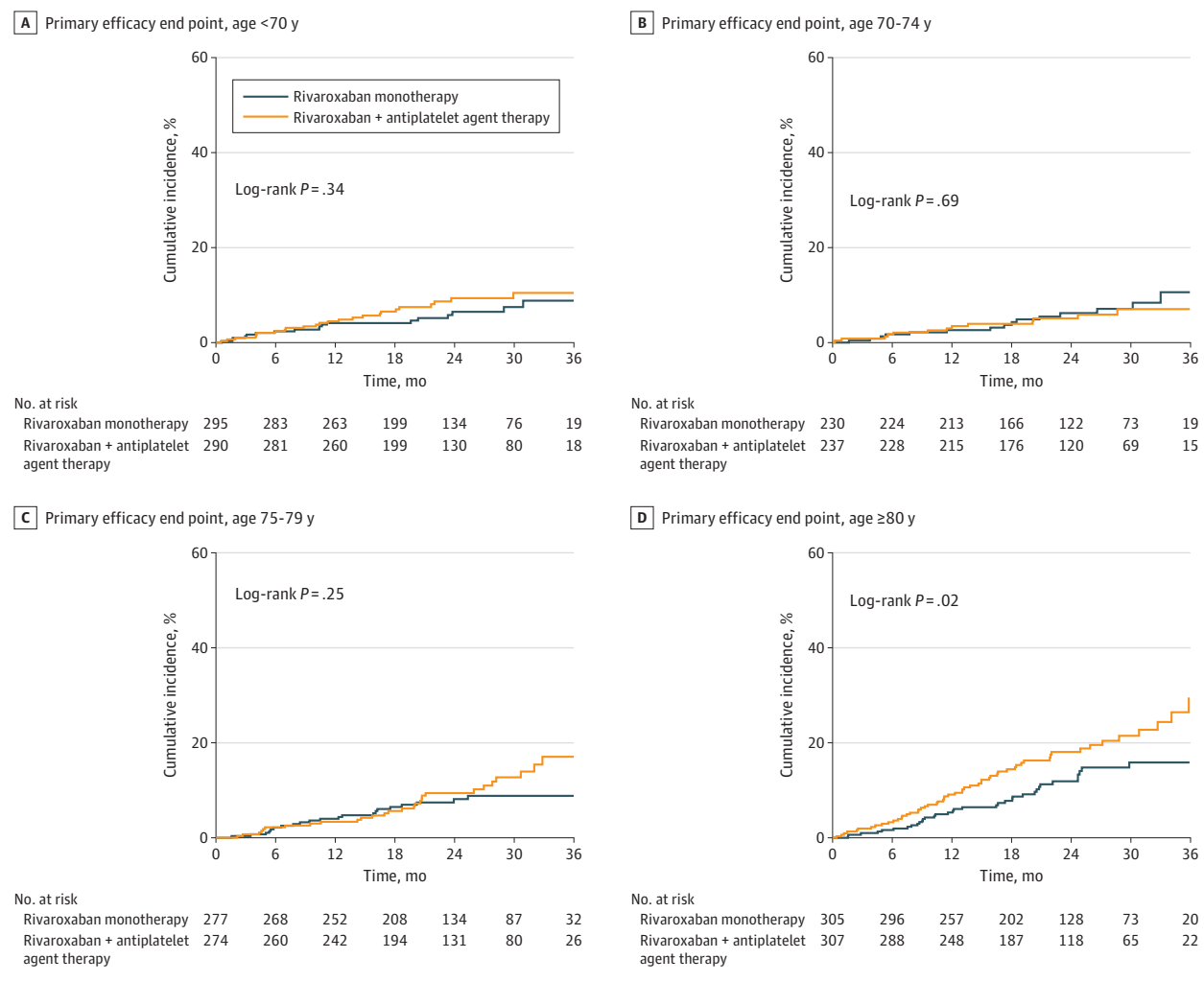
The incidence of the age-stratified primary composite efficacy end points is shown in **Figure 1** and **Figure 2**. In patients younger than 70 years, the primary efficacy end point occurred in 18 of 295 individuals (3.2%) taking rivaroxaban monotherapy vs 24 of 290 individuals (4.3%) taking rivaroxaban plus antiplatelet agent therapy. Among those aged 70 to 74 years, the primary efficacy end point occurred in 15 of 230 individuals (3.2%) taking rivaroxaban monotherapy vs 13 of 237 individuals (2.8%) taking rivaroxaban plus antiplatelet agent therapy. For those aged 75 to 79 years, the primary efficacy end point occurred in 21 of 277 individuals (3.8%) taking rivaroxaban monotherapy vs 28 of 274 individuals (5.3%) taking rivaroxaban plus antiplatelet agent therapy. For those 80 years or older, the primary efficacy end point occurred in 35 of 305 individuals (6.2%) taking rivaroxaban monotherapy vs 56 of 307 individuals (10.3%) taking rivaroxaban plus antiplatelet agent therapy. Patients in the groups taking rivaroxaban plus antiplatelet agent and rivaroxaban monotherapy had relatively similar risk of this end point in the age groups younger than 70 years (HR, 0.74; 95% CI, 0.40-1.37), 70 to 74 years (HR, 1.16; 95% CI, 0.55-2.45), and 75 to 79 years (HR, 0.72; 95% CI, 0.41-1.26), however, the group taking rivaroxaban monotherapy showed a lower risk than the group taking rivaroxaban plus antiplatelet agent therapy in those 80 years or older (HR, 0.61; 95% CI, 0.40-0.93). The effect of rivaroxaban monotherapy compared with that of combination therapy with rivaroxaban plus antiplatelet agent on

the primary efficacy end point was consistent across all age groups (*P* value for the interaction = .51).

Primary Safety Outcome

The age-stratified incidence of the primary safety end points are shown in **Figure 1** and **Figure 3**. In patients younger than 70 years, the primary safety end point occurred in 3 of 292 individuals (0.5%) taking rivaroxaban monotherapy vs 13 of 288 individuals (2.3%) taking rivaroxaban plus antiplatelet agent combination therapy. Among those aged 70 to 74 years, the primary safety end point occurred in 10 of 229 individuals (2.2%) taking rivaroxaban monotherapy vs 11 of 235 individuals (2.4%) taking rivaroxaban plus antiplatelet agent therapy; for those aged 75 to 79 years, the primary safety end point occurred in 6 of 276 individuals (1.1%) taking rivaroxaban monotherapy vs 11 of 272 individuals (2.1%) taking rivaroxaban plus antiplatelet agent therapy. For those 80 years or older, the primary safety end point occurred in 16 of 302 individuals (2.9%) taking rivaroxaban monotherapy vs 23 of 304 (4.3%) taking rivaroxaban plus antiplatelet agent therapy. Compared with those taking rivaroxaban plus antiplatelet agent, those taking rivaroxaban monotherapy had lower risk only those younger than 70 years (HR, 0.23; 95% CI, 0.06-0.79), whereas similar risk was observed in those aged 70 to 74 years, 75 to 79 years, and 80 years or older. Overall, the effect of rivaroxaban monotherapy compared with that of combination therapy with rivaroxaban plus antiplatelet agent on the primary safety end

Figure 2. Age Group–Stratified Cumulative Incidences of the Primary Efficacy End Point



point was consistent across the age spectrum (P value for interaction = .33).

Net MACE Plus Major Bleeding

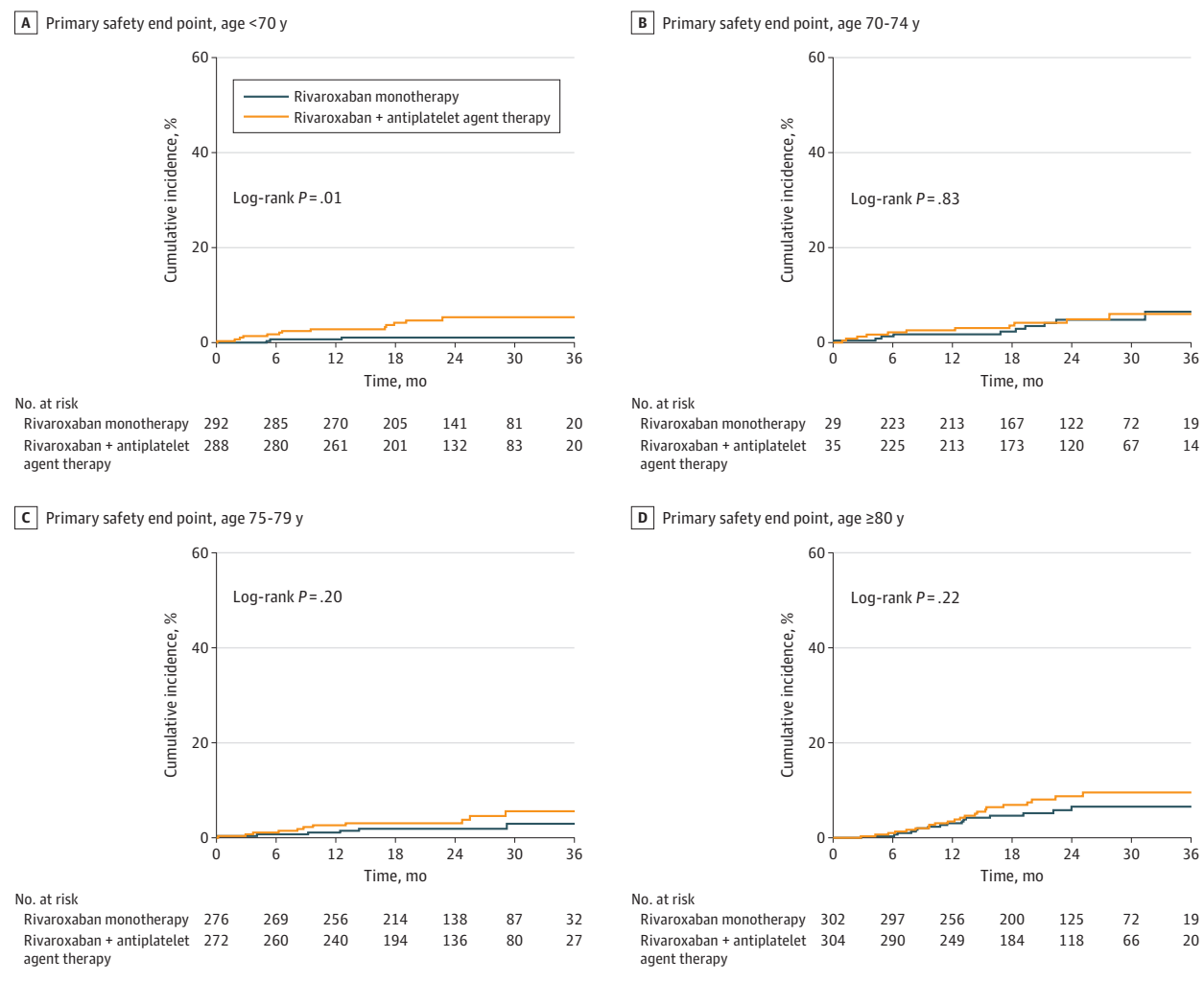
The age-stratified incidence rates of net MACE plus major bleeding are shown in Figure 1 and eFigure 1 in Supplement 1. In patients younger than 70 years, the net MACE plus major bleeding occurred in 15 of 295 individuals (2.6%) taking rivaroxaban monotherapy vs 26 of 290 individuals (4.7%) taking rivaroxaban plus antiplatelet agent therapy. Among those aged 70 to 74 years, the net MACE plus major bleeding occurred in 18 of 230 individuals (3.9%) taking rivaroxaban monotherapy vs 19 of 237 individuals (4.1%) taking rivaroxaban plus antiplatelet agent therapy. For those aged 75 to 79 years, the net MACE plus major bleeding occurred in 15 of 277 individuals (2.7%) taking rivaroxaban monotherapy vs 27 of 274 individuals (5.1%) taking rivaroxaban plus antiplatelet agent therapy; and for those 80 years and older, the net MACE plus major bleeding occurred in 36 of 305 individuals (6.4%) taking rivaroxaban monotherapy vs 59 of 307 individuals (10.9%) taking rivaroxaban plus antiplatelet agent therapy.

Compared with the rivaroxaban-antiplatelet group, the rivaroxaban-monotherapy group had relatively similar risk in those younger than 70 years and those aged 70 to 74 years but had lower risk in those aged 75 to 79 years and those 80 years or older (HR, 0.52; 95% CI, 0.28-0.98 and HR, 0.60; 95% CI, 0.40-0.91, respectively). Overall, the effect of rivaroxaban monotherapy compared with that of rivaroxaban plus antiplatelet agent therapy on net MACE plus major bleeding was consistent across the age spectrum (P for interaction = .56). In addition, eFigure 2 in Supplement 1 shows a direct comparison age group-stratified cumulative incidence of the primary efficacy end point (MACE) and primary safety end point (major bleeding).

Effect of Therapy With Age as a Continuous Variable

Figure 4 provides an alternative illustration of the effects of rivaroxaban monotherapy vs those of rivaroxaban plus antiplatelet agent therapy for the 3 previously described outcomes, derived using fractional polynomial analysis. The efficacy end point for rivaroxaban monotherapy was better in participants older than 75 years, whereas the safety end point was better in participants younger than 80 years. The overall

Figure 3. Age Group–Stratified Cumulative Incidences of the Primary Safety End Point



incidence of net MACE plus major bleeding was better for the rivaroxaban-monotherapy group across the age spectrum; however, the HR was significantly below 1 for participants older than 70 years.

Details of Participants Older Than 80 Years

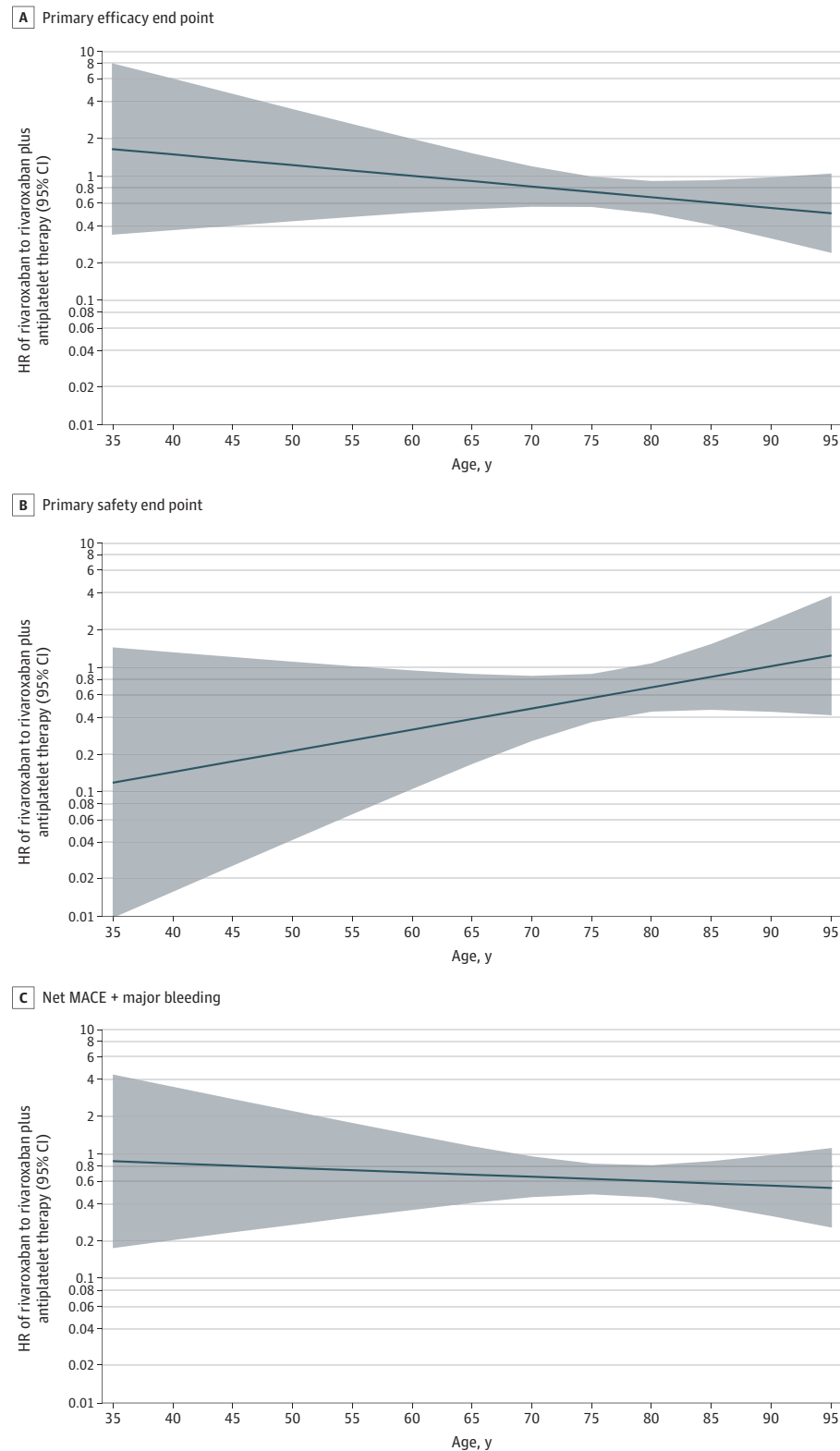
Details of the older participants (ie, aged >80 years) are shown in eTable 2 and eFigures 3 to 5 in Supplement 1. In patients aged 80 to 84 years, the primary efficacy end point occurred in 22 of 206 individuals (5.7%) taking rivaroxaban monotherapy vs 29 of 201 individuals (7.7%) taking rivaroxaban plus antiplatelet agent therapy, the primary safety end point occurred in 9 of 203 individuals (2.4%) taking rivaroxaban monotherapy vs 12 of 200 individuals (3.2%) taking rivaroxaban plus antiplatelet agent therapy, and the net MACE plus major bleeding occurred in 23 of 206 individuals (6.6%) taking rivaroxaban monotherapy vs 29 of 201 individuals (7.7%) taking rivaroxaban plus antiplatelet agent therapy. Among those older than 85 years, the primary efficacy end point occurred in 13 of 99 individuals (7.2%) taking rivaroxaban monotherapy vs 27 of 106 individuals (16.5%) taking rivaroxa-

ban plus antiplatelet agent therapy, the primary safety end point occurred in 7 of 99 individuals (3.9%) taking rivaroxaban monotherapy vs 11 of 104 individuals (6.7%) taking rivaroxaban plus antiplatelet agent therapy, and the net MACE plus major bleeding occurred in 13 of 99 individuals (7.2%) taking rivaroxaban monotherapy vs 30 of 106 individuals (18.3%) taking rivaroxaban plus antiplatelet agent therapy. The advantage of rivaroxaban monotherapy was evident, especially in participants older than 85 years, with a lower incidence of thrombotic events. This resulted in a net clinical benefit in these high-risk subsets.

Discussion

Results of this study revealed that rivaroxaban monotherapy exerted consistent effects, for both efficacy and safety, across the broad age groups in patients with AF and stable CAD more than 1 year after revascularization or angiographically confirmed CAD not requiring revascularization. In addition, the net efficacy of rivaroxaban monotherapy was consistent across all age groups;

Figure 4. Age Group–Stratified Effect of Rivaroxaban Monotherapy on Occurrences of the Primary Efficacy End Point, Primary Safety End Point, and Net Major Adverse Cardiovascular Event (MACE) Plus Major Bleeding



HR indicates hazard ratio.

however, efficacy was more pronounced in older patients, and safety was more pronounced in younger patients.

Although a report of the AFIRE trial was published in 2019 and may have influenced clinical practice to favor DOAC monotherapy, this treatment approach is not universally adopted. Because stable CAD in patients with AF is common globally and bleeding risk reduction is crucial, the AFIRE trial findings support anticoagulation monotherapy as a viable strategy. A recent meta-analysis of randomized clinical trials, including the AFIRE trial, supports the clinical benefit of oral anticoagulant monotherapy.¹⁷ In particular, older patients who have an elevated bleeding risk require careful assessment and close monitoring when considering DOAC monotherapy. Further research in more diverse populations remains necessary.

The average age of participants who were randomly assigned in the AFIRE group was 74 years, and 28% were 80 years or older. The global increase in the aging population has renewed interest in aging-associated diseases, such as AF. The efficacy of single-agent anticoagulant warfarin antithrombotic therapy in patients with stable CAD has been previously reported.¹⁸ Among patients with AF, a combination of antiplatelet agents and anticoagulants has been used after PCI based on the results of the Stent Anticoagulation Restenosis Study (STARS) trial¹⁹ and the Atrial Fibrillation Clopidogrel Trial With Irbesartan for Prevention of Vascular Event (ACTIVE W) trial.²⁰ Despite the trend toward shorter durations of combination therapy because of increased attention to bleeding complications, to our knowledge, no study has demonstrated the efficacy of a single DOAC in the greater than 1-year chronic phase. Approximately 65% of the participants in the AFIRE trial had stable CAD with prior coronary stenting. Matoba et al²¹ reported a consistent advantage of rivaroxaban monotherapy in this subset of the AFIRE trial population.

In recent years, evidence supporting the use of antithrombotic therapy for up to 1 year after PCI in patients with AF has increased.²²⁻²⁵ However, data guiding antithrombotic strategies beyond 12 months remain limited. In this context, therapy must carefully balance thrombotic and bleeding risks, particularly considering the distinct pathophysiology between the acute and chronic phases of CAD. In the stable phase, especially beyond 1 year after stenting, the thrombotic risk tends to decline, and prolonged antithrombotic therapy may lead to increased bleeding risk without a clear reduction in ischemic events. This trend has been reinforced by the evolution of drug-eluting stents, which have significantly lowered the risk of late stent thrombosis and shifted therapeutic priorities toward bleeding avoidance. Observational evidence from a nationwide Danish cohort study²⁶ suggested that anticoagulation alone may be an appropriate option for patients with AF beyond 1 year after PCI. This aligns with recent expert consensus-based guidelines, which recommended anticoagulation monotherapy despite limited randomized evidence available at the time.²⁷⁻²⁹ To address this clinical gap, the AFIRE trial was conducted as the first, to our knowledge, randomized clinical trial to demonstrate the efficacy and safety of rivaroxaban monotherapy in patients with AF and stable CAD more than 1 year after PCI. To the best of our knowledge, the current post hoc analysis of the AFIRE trial provides the first age-stratified evalua-

tion of this strategy, demonstrating that rivaroxaban monotherapy is both effective and safe, regardless of age.

Nonetheless, in older patients, it is important to remember that the patient background itself is a high-risk factor for not only bleeding but also thrombotic events. Specifically, kidney dysfunction and diabetes may contribute to a longer disease duration, and recent Japanese guidelines²⁷ noted an overlap in the risk of bleeding and thrombotic events. The results of the present study are of particular relevance for patients aged older than 85 years. In addition, the results of the detailed age-specific examination in the present study suggest that DOAC monotherapy may have a more pronounced effect in reducing thrombotic events than in reducing hemorrhagic events with increasing age. We assumed that a reduction in bleeding would reduce thrombotic events in our study population. Indeed, subanalyses of the AFIRE trial revealed an increase in the number of subsequent cardiovascular events in patients who experienced bleeding events.³⁰ The observed reduction in thrombotic rather than bleeding complications among older patients is noteworthy, and a potential underlying mechanism may exist. Although the present study did not clarify the factors involved, it is important to reaffirm that rivaroxaban monotherapy effectively reduces net adverse clinical events, which comprehensively include both thrombotic and hemorrhagic events, across a wide age range, even in older patients.

It is not clear if a class effect can be drawn from this study specifically. However, it is important to note that a recent meta-analysis¹⁷ and the Edoxaban vs Edoxaban With Antiplatelet Agent in Patients With Atrial Fibrillation and Chronic Stable Coronary Artery Disease (EPIC-CAD) trial,³¹ reporting that DOAC monotherapy led to a lower risk of MACE or major bleeding in patients with AF and stable CAD, indicated the absence of an interaction for age. Nonetheless, future DOAC monotherapy in patients with AF with stable CAD might require an individualized approach based on the clinical background of patients and their age. Indeed, there is a trend toward the identification of patients who are at high risk of bleeding and for reducing the use of antithrombotic drugs in individuals with CAD, particularly after PCI. Presently, guidelines advocate for the AFIRE trial-like regimen of DOAC monotherapy beyond 1-year post-PCI in patients with AF. A potential future consideration is how to shorten the duration of DOAC plus single antiplatelet agent treatment in patients with AF with stable CAD.

Limitations

The present study has some important limitations. First, the open-label design of the AFIRE trial may have increased the potential for bias.¹⁴ Second, as this was a post hoc analysis, the statistical analysis was largely underpowered to draw meaningful conclusions, and some differences in results between the 2 arms of the study could have been missed owing to the small sample size (type II statistical error). Third, the age categories chosen might have been arbitrary, although the number of patients in each group was almost balanced. Fourth, the choice of the antiplatelet agent was based on the decision of the treating physician and was not standardized. Fifth, the approved rivaroxaban dose in Japan is 15 mg once

daily, which is lower than the 20 mg dose commonly used in other countries. The pharmacokinetic equivalence observed in the AFIRE trial (where 15 mg rivaroxaban in Japanese patients is comparable with 20 mg globally) suggests that the findings may be relevant to other populations.³² However, further research in more diverse populations remains essential to ensure appropriate application across varying clinical settings. Sixth, as the Cox hazard model was not adjusted in the analysis, the inferences drawn from these intervals may not be reproducible. Therefore, the results of the present study should be interpreted cautiously.

Conclusions

Results of this post hoc analysis of the AFIRE randomized clinical trial reveal that rivaroxaban monotherapy reduced the risk of major cardiovascular events and major bleeding across the broad range of age in patients with AF and stable CAD. Possible age-related differences in trends, with more pronounced efficacy in older patients and more pronounced safety in younger patients, should be considered as hypothesis generating and require further research.

ARTICLE INFORMATION

Accepted for Publication: June 20, 2025.

Published Online: August 13, 2025.
doi:10.1001/jamacardio.2025.2611

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Obtained funding: Yasuda.
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Conflict of Interest Disclosures: Dr Yamaguchi reported receiving grants from Abbott, Boston Scientific, Terumo, Nippon Boehringer Ingelheim,

Bayer Yakuhin, Otsuka Pharmaceutical and personal fees from Abbott, Angs, Daiichi Sankyo, and Kowa outside the submitted work. Dr Hagiwara reported receiving grants from Bayer Yakuhin and Nippon Boehringer Ingelheim during the conduct of the study. Dr Kaikita reported receiving grants from Grants-in-Aid for Scientific Research from the Ministry of Education, Culture, Sports, Science, and Technology of Japan and personal fees from Bayer Yakuhin and Daiichi Sankyo outside the submitted work. Dr Akao reported receiving grants from Bayer Yakuhin and personal fees from Bayer Yakuhin, Daiichi Sankyo, and Bristol Myers Squibb outside the submitted work. Dr Ako reported receiving personal fees from Bayer outside the submitted work. Dr Matoba reported receiving grants from Bayer, Kowa, and Novartis outside the submitted work. Dr Nakamura reported receiving grants from Bayer Yakuhin and personal fees from Daiichi Sankyo, Bristol Myers Squibb, and Nippon Boehringer Ingelheim during the conduct of the study and endowments from Boston Scientific Japan, K.K., Asahi Intec Co Ltd Biotronik Japan Inc, Otsuka Medical Device Co Ltd, Kaneka Co, Terumo Corp, Nipro Co, and Japan Life Line Co Ltd. Dr Miyauchi reported receiving personal fees from Bayer during the conduct of the study. No other disclosures were reported.

Funding/Support: This study was supported by the Japan Cardiovascular Research Foundation through a contract with Bayer Yakuhin Ltd.

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Group Information: A complete list of the members of the AFIRE Investigators appears in [Supplement 2](#).

Data Sharing Statement: See [Supplement 3](#).

Additional Contributions: We thank Mebix, Tokyo, Japan, for their assistance in the management of data collection, storage, and analyses, and Editage, for English language editing and publication support. Financial compensation was received for these contributions.

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